Library Management Database Report

1. Introduction

Lantern Library requires an efficient and reliable database system to manage members, items, loans, fines, and inter-library operations. As the organisation expands its services, a structured relational design is essential to ensure accurate data handling and support future web-based access. This report follows the database system development lifecycle (DBSDL) to produce a fully normalised logical and physical models suitable for implementation in a multi-user environment.

2. Database Design

The Lantern Loans System is designed using semantic entity-relationship modelling and crow's foot notation following Barker/UML conventions to represent cardinality, optionality, and constraints clearly. Normalisation up to Third Normal Form (3NF) was applied to remove redundancy and ensure that each entity contains attributes dependent solely on its primary key.

The database design process progresses from requirements analysis to entity identification, relationship modelling, normalisation, and the development of logical and physical models.

2.1. Requirements Analysis (Assumptions and Design Choices)

Requirements were derived from the scenario, identified the need to store information about members, staff, items and their physical copies, loans, fines, publishers, categories, formats, libraries and inter-library loan activity, informing the identification of entities and relationships. To ensure realism, assumptions about library operations were made to guide the database design, with all resulting relationships, cardinalities, and ownership rules explicitly represented in the ER diagram.

Assumptions:

1. All **Members** recorded in the system are eligible to borrow items, and no membership status or suspension rules are modelled at the database level.
2. All **Staff** members are assumed to have equal operational responsibilities, and no role-based permissions are modelled in the system.
3. The **Biblio** entity represents the intellectual work rather than a physical instance, and physical availability is handled through **Item**.

4. **Biblio.Titles** are not assumed to be unique, and items are identified internally using a system-generated identifier.
5. Each physical **Item** is uniquely identified by a barcode.
6. Each **Item** is owned by exactly one **Library**, and ownership does not change in inter library processes.
7. Each **Loan** is associated with exactly one **Member**, and a **Member** may have multiple **Loans** over time.
8. An **Item** can appear only once in a single Loan.
9. **Fine** payment status is tracked using a simple paid/unpaid indicator.
10. **Fine** amounts are calculated externally based on overdue duration or policy rules, and only the final assessed amount is stored.
11. Temporal constraints that require cross-entity comparisons between **Loan** and **LoanItem** dates (LoanItem.ReturnDate>=Loan.LoanDate) are assumed to be enforced by application logic rather than database-level constraints, as such constraints cannot be reliably implemented using SQL CHECK constraints.

Design Choices:

1. **Biblio** entity stores exactly one **Publisher** and one **Category**. This reflects the fact that a **Publisher** and **Category** are bibliographic attributes that describe the edition of a work and is shared by all physical copies.
2. Each **Item** belongs to exactly one **Format** and describes physical copy rather than intellectual work. A single bibliographic record may have multiple **Items** with different **Formats**. Storing this at the **Item** level prevents duplication. **Format** is modelled as separate entity rather than an attribute in **Item** table, to avoid data redundancy, maintain normalization, and ensure clear and consistent classification, while avoiding unnecessary many-to-many relationships and associative entities. Although Format is currently a low-cardinality domain, storing it as a separate table does not constitute over-normalisation; instead, it provides a foundation for supporting new formats, format-specific policies, and activation control without requiring changes to the **Item** structure. This aligns with the Koha library system, where item types are configured entities with their own circulation rules and administrative settings.
3. Inter-library borrowing follows a workflow that is distinct from standard member loans and is therefore treated separately. Combining both processes within a single table would introduce NULL values, mix unrelated attributes, violate normalisation and reduce

semantic clarity. As a result, **InterLibraryLoan** is modelled as a separate entity from **Loan** to clearly represent these distinct business processes and to avoid the use of conditional or context-dependent attributes within a single entity. This separation is consistent with real systems such as Koha, which implements standard circulation and inter-library loans as distinct modules, thereby justifying this design choice.

4. **Loan** details are separated from individual borrowed items to avoid repeating groups and maintain normalization.
5. **LoanItem,** an associative entity**,** was introduced to resolve the many-to-many relationships between **Loan** and **Item** in compliance with normalization principles.
6. **ItemAuthor,** an associative entity, was introduced to resolve the many-to-many relationships between **Biblio** and **Author** in compliance with normalization principles.
7. **Fine** was associated with **LoanItem** rather than **Loan** to allow penalties to be applied to specific borrowed copies.
8. Address information is stored as single, composite attributes rather than as separate entities to reduce complexity, as detailed address breakdown is not required by the system scenario.
9. Emails are enforced as a unique attribute to reflect real-world uniqueness and to support identification, while avoiding its use as a primary key.
10. Barcode was defined as a unique attribute to reflect real-world identification of physical copies and prevent duplication.
11. Surrogate primary keys were used instead of natural keys to ensure stable entity identification and to simplify foreign-key referencing and indexing.
12. A surrogate primary key was combined with an alternate composite key in associative entities to simplify referencing while enforcing real-world uniqueness constraints.
13. Foreign keys to **Category**, **Format**, and **Publisher** were defined as NOT NULL to enforce complete bibliographic classification.
14. Two foreign keys in **Item** were introduced to represent the owner library and present location for item, ensuring clear role identification.
15. Two foreign keys in **InterLibraryLoan** were used to represent the source and destination libraries, ensuring clear role identification.
16. CHECK constraints were used on the **Fine.Amount** and **Fine.PaidStatus** attributes to ensure fines have valid monetary values and enforce permitted payment states at the database level.

17. CHECK constraints were applied to **Status**, **DateRequested**, **DateReceived**, **DueDate**, and **DateReturned** attributes in **Loan** and **InterLibraryLoan** tables to enforce valid lifecycle states and basic intra-entity date ordering.
18. Table-level CHECK constraints were applied only where business rules span multiple attributes, specifically in the **Loan** and **InterLibraryLoan** tables. These constraints enforce valid date sequencing and ensure logical relationships such as preventing inter-library loans within the same library. Other entities contain independent attributes and therefore rely solely on column-level constraints.

2.2. Entity Identification

Entities were determined through the analysis of the scenario and the mapping of important nouns such as Member, Item, Staff, Loan, and Library to entity types. On the other hand, the operational verbs like borrow, process, issue fines, and transfer helped determine how these entities relate to one another.

The entities used in the database includes **Member, Staff, Biblio, Item, Loan, Fine, Category, Format, Publisher, Library,** and **InterLibraryLoan**. Although the scenario does not explicitly mention authors, an **Author** entity was added to make the system more realistic. Additionally, two associative entities—**LoanItem** and **ItemAuthor**—were introduced during normalisation to resolve many-to-many relationships. Together, these entities provide full coverage for managing loans, item circulation, fines, and inter-library transfers.

2.3. Relationships

Most relationships in the Lantern Loans System are straightforward one-to-many structures, such as Member–Loan, Biblio–Item, Library–Staff, and therefore require no further justification.

Several relationships required additional consideration due to their structural or semantic complexity:

- Loan – Item (M:N). This many-to-many relationships are resolved using the associative entity **LoanItem**.
- Biblio – Author (M:N). These relationships are represented using the **ItemAuthor** associative entity.

- Item – Lybrary (M:1). Each **Item** is associated with exactly one owning library and one current holding location. These roles are represented using the role-named foreign keys **HomeLibraryID** and **HoldingLibraryID**.
- Library – InterLibraryLoan (1:M). Each **InterLibraryLoan** references one source library and one destination library. These roles are represented using the role-named foreign keys **FromLibraryID** and **ToLibraryID**.

2.4. Normalization

Normalisation was applied systematically to ensure that the Lantern Loans System database is free from redundancy, updating anomalies, and inconsistent dependencies. The process followed the standard three-stage decomposition approach, progressing from unnormalised data to a fully normalised set of entities in Third Normal Form (3NF).

2.4.1 First Normal Form (1NF)

First Normal Form (1NF) requires that all attributes be atomic and that no repeating groups exist within relations. In the Lantern Loans System, multi-valued and repeating data were removed by decomposing Loan–ItemCopy and Item–Author structures into associative entities. As a result, all entities contain only atomic attributes and satisfy 1NF requirements.

2.4.2 Second Normal Form (2NF)

A relation is in 2NF when it is in 1NF and all non-key attributes depend on the whole primary key. Most entities use a single-attribute primary key and therefore already satisfy 2NF. The associative entities LoanItem and ItemAuthor use surrogate keys, ensuring that all attributes depend on the full primary key and eliminating partial dependencies. A surrogate key cannot create partial dependencies because it is a single attribute. Any table using a surrogate key is automatically in 2NF.

2.4.3 Third Normal Form (3NF)

A database is in Third Normal Form when it is in 2NF and contains no transitive dependencies. All non-key attributes in the Lantern Loans System depend solely on their respective primary keys. Descriptive data such as category names, publisher details, and library information are stored in separate relations, preventing non-key dependencies and ensuring compliance with 3NF.

2.5 Logical Model

The logical model represents the structure of the Lantern Loans System independently of any specific DBMS. It extends the conceptual model into a fully normalised schema with clearly defined entities, keys, and constraints.

2.6 Physical Model

The physical model implements the logical design as a relational schema suitable for implementation in a relational DBMS. It defines data types, primary keys, foreign keys, alternate keys, and integrity constraints for all tables.

Surrogate primary keys are used consistently to support efficient indexing and foreign-key referencing. Alternate keys are applied where required to enforce business-level uniqueness.

Data types are selected according to attribute purpose, including INT for identifiers, VARCHAR for descriptive attributes, DATE for temporal data, and DECIMAL for monetary values. CHECK constraints are used to enforce valid domain values for attributes such as item status and fine payment state.

Overall, the physical model provides a normalised and implementable database schema that maintains referential integrity and supports reliable operation of the Lantern Loans System.